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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Task Description:	The general objective of this project is to determine how animal cells detect gravitational forces, and how gravitational forces affect cell behavior such as migration. Changes in gravity cause dramatic changes in mechanical load and rheological properties in various parts of the body, therefore it is crucial to understand how these factors affect single cells. We have probed living fibroblasts with manipulations that simulate the effects of gravity, such as pushing or pulling forces and changes in substrate flexibility or topography. The responses have been studied by microscope imaging of cell shape and migration, cytoskeletal structures, and traction forces. In addition, cytoskeletal structures and intracellular signaling pathways have been manipulated with chemical, physical, and molecular biological means, and the impact on mechanosensing determined. These experiments have lead to a better understanding of both the responses of cells to gravity-related signals, and the underlying mechanisms... The general objective of this project is to determine		

	how animal cells detect gravitational forces, and how gravitational forces affect cell behavior such as migration. Many profound physiological effects associated with space flights are likely caused by reduced weight load and shifts in body fluid distribution, which in turn cause drastic changes in tissue mechanical properties and in mechanical input into cells. Thus to understand the effects of gravity, it is crucial to determine at a basic science level how animal cells respond to such physical signals as substrate flexibility and mechanical forces. The proposed experiments address these questions directly. The results will provide a foundation for understanding and treating health problems associated with space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cellular mechanosensing affects a number of physiological and pathological processes, from embryonic development, wounding healing, to cancer invasion. This project will provide a foundation for understanding these processes and for treating related diseases.
Task Progress:	1. Regulation of phagocytosis by mechanical signals. Phagocytosis is an actin-based process used by macrophages to clear particles greater than 0.5 mm in diameter. In addition to its role in immunological responses, phagocytosis is also necessary for tissue remodeling and repair. To prevent catastrophic autoimmune reactions, phagocytosis must be tightly regulated. It is commonly assumed that the recognition/selection of phagocytic targets is based solely upon receptor-ligand binding. However, we suspect that physical factors also play an important role. We prepared micron-sized particles of identical chemical properties but different rigidity, using polyacrylamide crosslinked with different concentrations of bis-acrylamide. The particles were opsonized and fed to mouse bone marrow derived macrophages. While the particles can adhere to the cell surface irrespective of their rigidity, there was a strong preference for macrophages to engulf rigid objects. Furthermore, phagocytosis of soft particles can be stimulated with the microinjection of constitutively active Rac1 but not RhoA, and with lysophosphatidic acid, an agent known to activate the small GTP-binding proteins of the Rho family. These data suggest a Rac1-dependent mechanosensory mechanism for phagocytosis. Changes in tissue rigidity under microgravity conditions may play a role in autoimmune reactions during space flight. 2. Regulation of cell shape and migration by environmental topography. Fibroblasts in 2D cultures differ dramatically in behavior from those in the 3D environment of a multicellular organism. However the basis of this disparity is unknown. A key difference is the spatial arrangement of anchored extracellular matrix (ECM) receptors only to the ventral surface in 2D cultures, and throughout the entire surface in 3D cultures. Therefore, we asked if changing the topography of ECM receptor anchorage alone could invoke a morphological response. To present anchored fibronectin or collagen to dorsal cell surfaces, we cultured cells on flexible polyacrylamide substrates and applied a similar sheet to the top surface of the cell, embedding cells within a sandwich. Fibroblasts quickly changed from a well-spread polygonal morphology on 2D surfaces into a bipolar or stellate morphology, similar to fibroblasts in vivo. Cells in this environment lacked lamellipodia and large actin bundles, and formed small focal adhesions only near focused sites of protrusion. Imaging with fluorescent calcium indicators indicated that there was an increase in calcium upon the engagement of dorsal receptors. In addition, calpain-null cells failed to respond to dorsal substrate, suggesting that one of the effectors of calcium may be the calcium-dependent protease calpain at focal adhesions. The response also appeared to involve small GTPases, as cells expressing constitutively active small GTPase Rac remained discoid in shape upon the application of dorsal substrate. These results suggest that spatial distribution of anchored ECM receptors plays a critical role in determining cell shape and behavior, which may then affect diverse activities including gene expression, growth, and differentiation as shown in numerous previous studies. An aspect relevant to microgravity is that the response is dependent on the rigidity of the dorsal substrate. While most cells in the body existed in a homogenous 3D environment, a transition to 2D-like asymmetry occurs during wound healing. Reduced gravity may affect the mechanical asymmetric thereby disrupting the wound healing responses. 3. Responses of microtubules to mechanical forces While membrane receptors and channels represent the main candidates in mediating the responses to mechanical signals, a more intriguing mechanism involves the cytoskeleton. Mechanical forces may induce changes in cytoskeletal organization, which may then affect a number of important cellular functions. To test a possible mechanosensing mechanism that involve microtubule assembly and guidance toward target structures, we have applied stress to cells either by pushing the body of B16 melanoma cells with a needle or by stretching a flexible substrate. We found that while the lamella region behind rapidly protruding lamellipodia typically showed few microtubules, application of stress caused an approximately three-fold increase in the number of microtubules and a dramatic increase in the number of plus ends of microtubules in this region. The response was accompanied by an increase in actin filament bundles extending from the cell center to the periphery. Similar responses were observed by stretching flexible substrates underneath cells, so that after 1 minute of tension application there was an average of 2.5-fold increase in the number of microtubules in anterior lamella zones. Microtubules may respond directly to applied forces, or indirectly to changes in signaling molecules such as calcium and phosphorylation. Due to the extensive presence of the cytoskeletal network, these structures may provide a powerful mechanism to induce responses to mechanical signals in large regions of the cell. 4. Involvement of myosin II in cell shape, migration and mechanosensing Adherent cells cultured on flat, homogenous surfaces typically maintain an intact cell body with a polygonal or fan shape, despite active migration and strong mechanical interactions with the substrate. In addition to the surface membrane, an active mechanism may be involved in maintaining the shape and integrity of the cell body, particularly in the body where cells encounter complex topographic patterns of guidance cues. To detect if there is an active mechanism that constrains cell shape, we plated NIH 3T3 fibroblasts on ring-patterned substrates 8-22 microns in width and 53-140 microns in outer diameter, prepared by photolithography. Untreated cells have a limited angular span, encompassing an average of 105 degrees around the ring, even though these cells were able to cover a much larger surface when plated on flat surfaces of the same material. Treatment of 3T3 cells with the myosin II inhibitor, blebbistatin, induced a striking increase in the bending ability, causing cells to cover more than 60% of the ring. Inhibition of the Rho-dependent kinase with Y-27632 caused a similar increase in the angular span. These results suggest that cell shape is controlled not only by the passive constraint of surface membrane but also by an active

	mechanism driven by myosin II-mediated contractility under the regulation of the small GTPase Rho. This mechanism allows the cell to maintain its integrity while navigating through a complex physiological environment. In addition, as our previous studies suggest that traction forces are regulated by mechanical signals such as substrate rigidity, we suspect that microgravity may cause a decrease in mechanical signals in some parts of the body and weaken myosin II-mediated contractility, thereby compromising the integrity of the cell. We have further addressed the function of one of the isoforms of myosin II, myosin IIB, by analyzing the movement and mechanical characteristics of fibroblasts where this protein has been ablated by gene disruption. Myosin IIB null cells displayed multiple unstable and disorganized protrusions, although they were still able to generate a large fraction of traction forces when cultured on flexible polyacrylamide substrates. However, the traction forces were highly disorganized relative to the direction of cell migration. Analysis of cell migration pattern indicated an increase in speed and decrease in persistence. These defects were likely responsible for the defects in directional movements, as detected with Boyden chambers coated on one side with matrix proteins. In addition, unlike control cells, mutant cells failed to respond to mechanical signals such as compressing forces and changes in substrate rigidity. These results suggest that myosin IIB is involved not in propelling but in directing the cell movement, by coordinating protrusive activities and stabilizing the cell polarity.
Bibliography Type:	Description: (Last Updated: 06/16/2020)
Articles in Peer-reviewed Journals	Lo CM, Buxton DB, Chua GC, Dembo M, Adelstein RS, Wang YL. "Nonmuscle myosin IIB is involved in the guidance of fibroblast migration." Mol Biol Cell. 2004 Mar;15(3):982-9. PMID: 14699073 , Mar-2004
Articles in Peer-reviewed Journals	Beningo KA, Dembo M, Wang YL. "Responses of fibroblasts to anchorage of dorsal extracellular matrix receptors." Proc Natl Acad Sci U S A. 2004 Dec 28;101(52):18024-9. PMID: 15601776 , Dec-2004